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Title : Ai intern CaseStudy

Demo link :

https://drive.google.com/file/d/1DXF_LJqCNxiB1tXqZn0UkdqMLI283goD/view?usp=sharing

STUDENTS INTRODUCTION SCORER

1. ABSTRACT

This project presents a lightweight web-based tool designed to evaluate a student's spoken introduction using a predefined rubric. The application accepts a text transcript of a student's introduction and produces a detailed scoring breakdown along with constructive feedback. The system uses rule-based evaluation combining keyword coverage and length analysis for each rubric criterion. The tool is built using Python and Streamlit, providing a clean and responsive interface for easy use by teachers or evaluators. The solution aligns with the requirements given by Nirmaan AI for their internship case study and demonstrates practical problem-solving, rubric structuring, and product-oriented development.

2. INTRODUCTION

Spoken introductions are an essential communication skill for students. Evaluating these introductions manually can be inconsistent and time-consuming. The goal of this project is to automate this process by creating a simple and user-friendly web application that evaluates text-based student introductions using a rubric.

The rubric describes what an ideal introduction should contain, including:

- Basic self-introduction
- Family and background
- Interests and hobbies
- Academic interests and future goals

This project converts that rubric into an interactive scoring tool.

3. PROBLEM STATEMENT

The Nirmaan internship case study required the development of:

“A small tool that evaluates a student’s spoken introduction using the given transcript.”

The tool should:

- Take a transcript as input
- Use a rubric to evaluate the content
- Produce scores and feedback
- Be simple, efficient, and easy to use
- No advanced machine learning is required a logical, well-designed rule-based approach is acceptable and encouraged.

4. PROJECT OBJECTIVE

The main objectives are:

- Understand the rubric from the provided Excel file
- Build a functional scoring system around the rubric
- Design a clean and modern interface
- Make the system intuitive and easy for evaluators
- Provide actionable feedback with transparency
- Maintain low dependency and simple deployment

5. REQUIREMENTS

5.1 Software Requirements

- Python 3.x
- Visual Studio
- Streamlit
- Pandas
- NumPy

5.2 Hardware Requirements

- Basic laptop or PC
- Any modern browser like chrome, firefox

5.3 Input Requirements

The system accepts a text transcript of a student introduction converted from voice or typed manually.

Sample text:

Hello everyone, myself Muskan, studying in class 8th B section from Christ Public School. I am 13 years old. I live with my family. There are 3 people in my family, me, my mother and my father. One special thing about my family is that they are very kind hearted to everyone and soft spoken. One thing I really enjoy is play, playing cricket and taking wickets. A fun fact about me is that I see in mirror and talk by myself. One thing people don't know about me is that I once stole a toy from one of my cousin. My favorite subject is science because it is very interesting. Through science I can explore the whole world and make the discoveries and improve the lives of others.

Thank you for listening.

5.4 Output Requirements

The system displays:

- ✓ Total word count
- ✓ Per-criterion scores
- ✓ Keyword coverage
- ✓ Length evaluation
- ✓ Overall weighted score
- ✓ Constructive feedback

6.PROPOSED SYSTEM:

The proposed system is a simple and practical web application designed to automatically evaluate a student's self-introduction based on a structured rubric. Instead of manually listening to every introduction and scoring it individually, this tool provides a faster and more consistent way to assess students using text input.

The system is built using Python and Streamlit, which together make it lightweight, easy to use, and suitable for both teachers and evaluators. The scoring logic follows a rule-based approach, where each part of the introduction is analyzed based on important keywords and the length of the response. This ensures that the evaluation remains transparent, simple to understand, and closely aligned with the rubric provided in the Nirmaan internship case study.

The application allows the user to paste the transcript of a student's introduction and instantly receive a structured result. The final output includes the total word count, detailed scores for each rubric category, and constructive feedback that can help the student improve their communication skills. The rubric itself is embedded inside the system so that the app does not rely on external Excel files during execution.

The main components used in the proposed system include:

- **Python** as the core programming language
- **Streamlit** for the user interface
- **Pandas and NumPy** for text processing and score calculations

- **Rule-based scoring logic** based on rubric keywords and expected length

The interface is intentionally kept clean and simple. The main section contains the input area and scoring results, while the sidebar displays the rubric summary and developer credit. This layout ensures that users can clearly understand what the system is evaluating without any confusion.

Overall, the proposed system offers a practical and user-friendly solution for evaluating student introductions. It automates the grading process, reduces evaluator workload, and provides consistent scoring every time. The design also allows for easy future expansion, such as adding speech-to-text input or more advanced semantic scoring if needed.

7.MODULES:

Rubric Module

- Hardcoded rubric criteria
- Weights, keywords, descriptions

Text Processing Module

- Cleans and normalizes input
- Counts words

Rule-Based Scoring Engine

- Keyword matching
- Length comparison
- Weighted scoring

User Interface Module

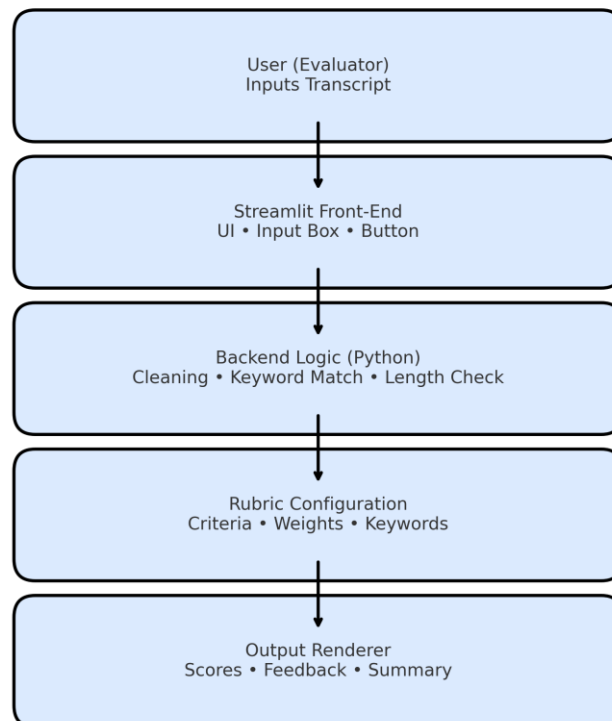
- Streamlit-based UI
- Sidebar rubric display

- Results section with metrics and feedback

8. RUBRICS DESCRIPTION:

Criterion	Description	Weight
Basic Self-Introduction	Name, class, school	0.25
Family & Background	Family details, personal info	0.25
Interests & Hobbies	What student likes or enjoys	0.20
Academic Interests & Goals	Favourite subject, goals	0.30

9.SYSTEM ARCHITECTRE

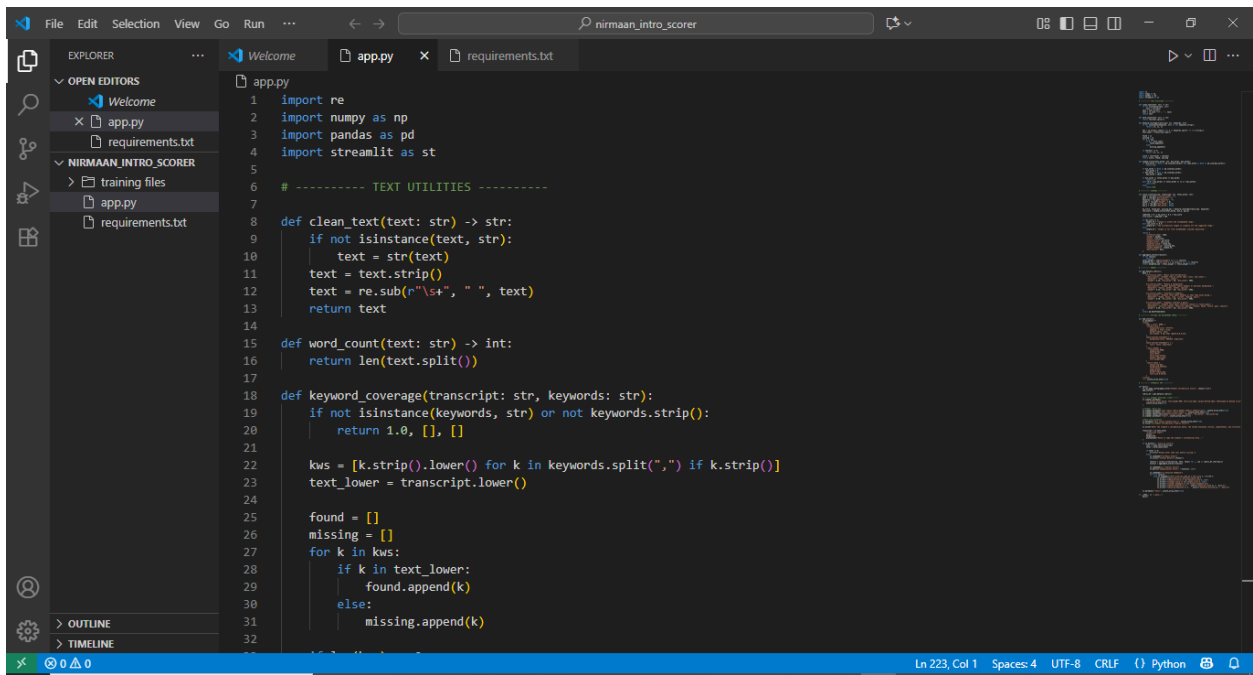


10. WORKING OF THE SYSTEM

Step-by-Step Flow:

1. User enters or pastes transcript
2. System cleans and analyzes text
3. For each rubric criterion:
 - Keyword match
 - Length validation
 - Score calculation
4. Scores are weighted
5. Overall score computed
6. Feedback displayed

11. SAMPLE SCREENSHOT :



```
File Edit Selection View Go Run ... Welcome app.py requirements.txt
EXPLORER
  OPEN EDITORS
    Welcome
    app.py
    requirements.txt
  NIRMAAN_INTRO_SCORER
    training files
    app.py
    requirements.txt
  OUTLINE
  TIMELINE
  0 0 0

1 import re
2 import numpy as np
3 import pandas as pd
4 import streamlit as st
5
6 # ----- TEXT UTILITIES -----
7
8 def clean_text(text: str) -> str:
9     if not isinstance(text, str):
10         text = str(text)
11     text = text.strip()
12     text = re.sub(r"\s+", " ", text)
13     return text
14
15 def word_count(text: str) -> int:
16     return len(text.split())
17
18 def keyword_coverage(transcript: str, keywords: str):
19     if not isinstance(keywords, str) or not keywords.strip():
20         return 1.0, [], []
21
22     kws = [k.strip().lower() for k in keywords.split(",") if k.strip()]
23     text_lower = transcript.lower()
24
25     found = []
26     missing = []
27     for k in kws:
28         if k in text_lower:
29             found.append(k)
30         else:
31             missing.append(k)
```

Home page :

Developed by Renuka A

	criterion_name	weight	min_words	max_words
0	Basic self-introduction	0.2500	30	200
1	Family & background	0.2500	20	200
2	Interests & hobbies	0.2000	20	200
3	Academic interest & goals	0.3000	20	200

Student Introduction – Rubric Scorer

Enter the student's introduction below. The system evaluates clarity, completeness, and structure based on the rubric.

Transcript Input

Paste or type the student's introduction here...

Score my intro

User enter input:

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	criterion_name	weight	min_words	max_words
0	Basic self-introduction	0.2500	30	200
1	Family & background	0.2500	20	200
2	Interests & hobbies	0.2000	20	200
3	Academic interest & goals	0.3000	20	200

Student Introduction – Rubric Scorer

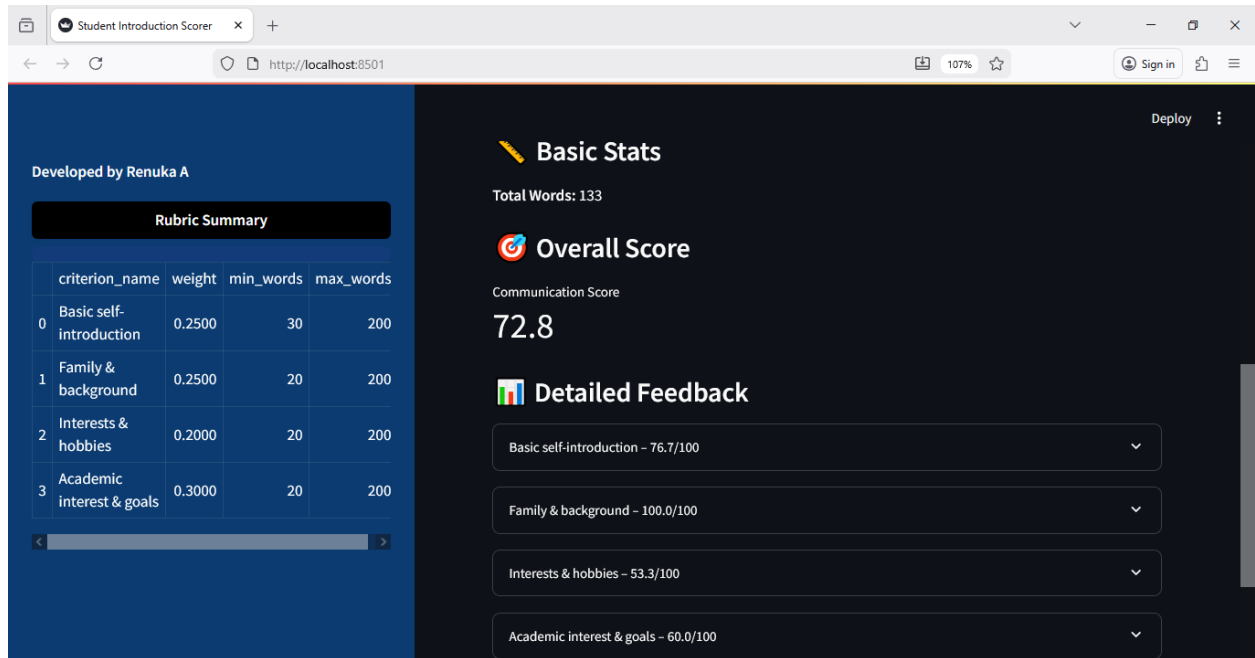
Enter the student's introduction below. The system evaluates clarity, completeness, and structure based on the rubric.

Transcript Input

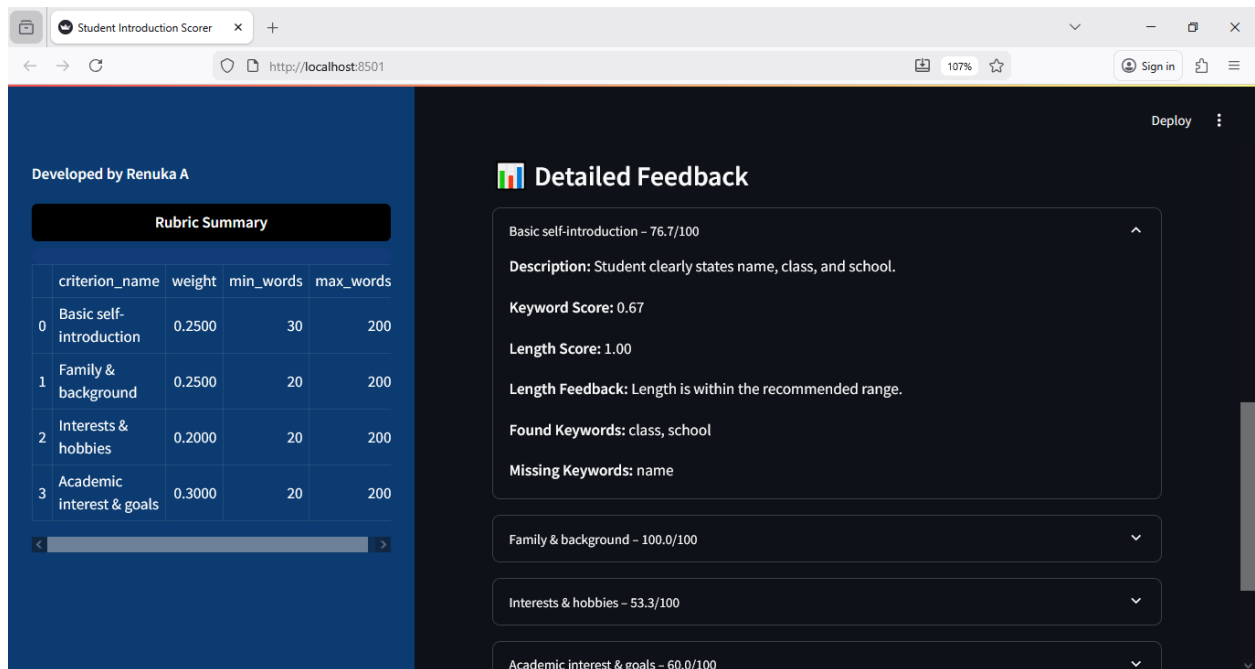
Hello everyone, myself Muskan, studying in class 8th B section from Christ Public School. I am 13 years old. I live with my family. There are 3 people in my family, me, my mother and my father. One special thing about my family is that they are very kind hearted to everyone and soft spoken. One thing I really enjoy is play, playing cricket and taking wickets. A fun fact about me is that I see in mirror and talk by myself. One thing people don't know about me is that I once stole a toy from one of my cousin. My favorite subject is science because it is very interesting. Through science I can explore the whole world and make the discoveries and improve the lives of others.

Score my intro

Score:



Score Results :



Student Introduction Scorer

← → ↺

http://localhost:8501

107%

Sign in

⌵

Developed by Renuka A

Rubric Summary

	criterion_name	weight	min_words	max_words
0	Basic self-introduction	0.2500	30	200
1	Family & background	0.2500	20	200
2	Interests & hobbies	0.2000	20	200
3	Academic interest & goals	0.3000	20	200

< >

Deploy ⋮

Family & background – 100.0/100

Description: Student mentions family members or personal background.

Keyword Score: 1.00

Length Score: 1.00

Length Feedback: Length is within the recommended range.

Found Keywords: family, mother, father, special

Missing Keywords: None

Interests & hobbies – 53.3/100

Academic interest & goals – 60.0/100

Student Introduction Scorer

← → ↺

http://localhost:8501

107%

Sign in

⌵

Developed by Renuka A

Rubric Summary

	criterion_name	weight	min_words	max_words
0	Basic self-introduction	0.2500	30	200
1	Family & background	0.2500	20	200
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3	Academic interest & goals	0.3000	20	200

< >

Deploy ⋮

Interests & hobbies – 53.3/100

Description: Student talks about hobbies or what they enjoy doing.

Keyword Score: 0.33

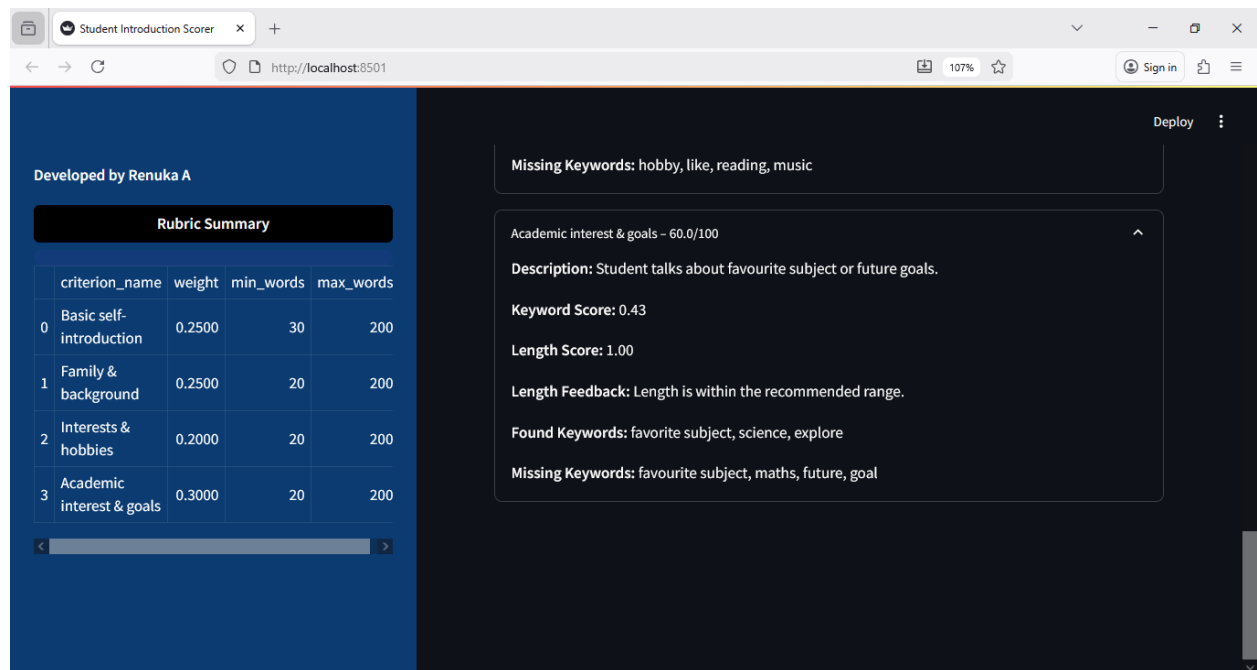
Length Score: 1.00

Length Feedback: Length is within the recommended range.

Found Keywords: enjoy, cricket

Missing Keywords: hobby, like, reading, music

Academic interest & goals – 60.0/100



12.CONCLUSION:

This project successfully fulfills the requirements of the Nirmaan AI internship case study by transforming a rubric into a functional scoring tool. The system is user-friendly, lightweight, and demonstrates strong problem-solving and product-thinking skills. It serves as a foundation for more advanced AI-driven evaluation tools in the future.